

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2023

Programme: BE

Full Marks: 100

Course: Digital Signal Analysis and Processing

Pass Marks: 45

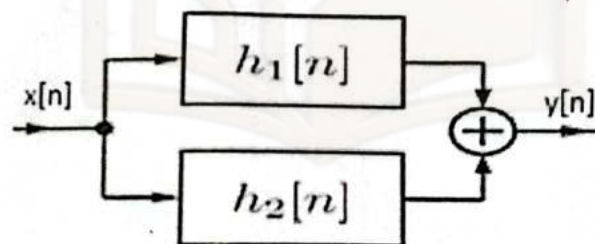
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) A digital communication link carries binary coded words representing samples of an input signal $x_a(t) = 5 \cos 600\pi t + 7 \cos 800\pi t$. The link is operated at 1000 bits/sec and each input sample is quantized into 1024 different voltage levels. 7
- i. What is the sampling frequency and folding frequency?
 - ii. What is the Nyquist rate for the signal $x_a(t)$?
 - iii. What are the frequencies in the resulting discrete time signal $x[n]$?
 - iv. What is aliasing effect?
- b) Two subsystems $h_1[n]$ and $h_2[n]$ are interconnected as shown in the block diagram. Determine the response of the system if; $h_1[n] = \{1, 4, 2\}$ and $h_2[n] = \{2, 3, 1\}$, when excited by input $x[n] = \{2, 4\}$. 8



2. a) Determine which signals are periodic and compute fundamental period. 8
- i. $\cos\left(\frac{\pi n^2}{8}\right)$
 - ii. $\sin\left(\frac{3\pi}{2}\right) \cos\left(\frac{5\pi n}{4}\right)$
- b) Define z-transform and Region of Convergence. Find z-transform of following signal $x[n] = a^n u[n]$. 7
3. a) Why do we need DFT when we have DTFT? Determine the circular convolution of the sequence: $x_1(n) = \{4, 3, 2, 1\}$ and $x_2(n) = \{8, 7, 6, 5\}$ 7
- b) Determine the direct form realization of the following difference equation. 8

$$2y(n) + y(n-1) - 4y(n-3) = x(n) + 3x(n-1)$$

Compare FIR and IIR

4. a) Determine the cascade and parallel realization of the discrete time system described by differential equation. 8

$$y[n] = -\frac{3}{4} y[n-1] + \frac{1}{4} y[n-2] + x[n] + \frac{1}{2} x[n-1]$$

- b) Obtain the lattice ladder structure of the discrete time system described by the differential equation 7

$$y[n] = -\frac{3}{4} y[n-1] + \frac{1}{4} y[n-2] + x[n] + \frac{1}{2} x[n-1]$$

Also check the stability of the filter

5. a) Design a digital low pass Butterworth filter by applying bilinear transformation technique for the given specifications. 7

Pass band edge = 120Hz

Pass band attenuation = 1dB

Stop band edge = 170Hz

Stop band attenuation = 16 dB Assume sampling frequency of 512 Hz

- b) Obtain $H(z)$ using the impulse invariant techniques for an analog system function which is given by: 8

$$H_a(S) = \frac{1}{(S+0.5)(S^2+0.5S+2)}$$

6. a) Design a linear FIR filter using Kaiser window to meet the following specifications: 8

$$0.99 \leq |H(e^{jw})| \leq 1.01; \text{ for } 0 \leq |w| \leq 0.19\pi$$

$$|H(e^{jw})| \leq 0.01; \text{ for } 0.21\pi \leq |w| \leq \pi$$

- b) By using Hanning window, design a low pass filter to approximate the ideal response given by: 7

$$H(e^{jw}) = \begin{cases} 1 & \text{for } -\frac{\pi}{6} \leq w \leq \frac{\pi}{6} \\ 0 & \text{otherwise} \end{cases}$$

Use the filter length of $M=9$ for your design

7. Write short notes on: (Any two)

2×5

- Sampling and quantization of analog signal,
- BIBO Stable
- Frequency Sampling